

Endomembranes

Current 47-page faculty deck primary - official Week 4 objectives integrated - protein targeting, ER quality control, vesicle coats, Rab/SNARE fusion, lysosomal sorting, secretion, and endocytosis

5 official objectives

SRP/translocon

COPI/COPII/clathrin

Rab + SNARE

M6P

CFTR

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Define signal sequence and describe how signal sequences sort proteins to cellular organelles.
2. Describe ER protein-folding quality control and the fate of misfolded proteins.
3. Describe the molecular mechanism of vesicle docking and membrane fusion.
4. Compare constitutive and regulated secretory pathways from Golgi to plasma membrane.
5. Explain cargo transport into cells and between organelles, the roles of coat proteins, and diseases caused by trafficking defects.

ASSIGNED SCHOLARRX

Endoplasmic Reticulum & Golgi Apparatus

Eukaryotic cells are highly compartmentalized, so hydrophilic macromolecules cannot simply diffuse through lipid bilayers. Many proteins carry **signal sequences** that are recognized by receptors and direct them to the correct organelle.

Destination	Transport logic	Lecture emphasis
Nucleus	Gated transport	Handled in the nucleus lecture
Mitochondrion	Transmembrane, usually post-translational	N-terminal targeting signal; chaperones keep precursor unfolded
ER	Usually co-translational	Hydrophobic signal peptide + SRP/translocon

No signal = default matters: without a recognized targeting signal, a newly made soluble protein generally remains in the cytosol rather than magically entering an organelle.

hydrophobic signal emerges

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SRP binds signal + ribosome

->

translation pauses

->

SRP receptor docks at ER

->

translocon opens

->

translation resumes into ER

SRP is a ribonucleoprotein and GTP-binding factor. Signal peptidase can cleave the N-terminal signal peptide. Membrane proteins can use start-transfer and **stop-transfer** sequences to establish topology.

ER modifications: N-linked glycosylation begins in the ER. Oligosaccharyl transferase adds an oligosaccharide to asparagine residues in the ER lumen, helping folding and quality control.

Slides 25-27

The ER tests conformation before allowing cargo to leave

LO 2

fold with chaperones

->

glycan trimming acts as timer

->

persistent misfolding

->

retrotranslocation

->

ubiquitin

->

proteasome

Accumulation of unfolded proteins activates the **unfolded protein response (UPR)**: global protein synthesis is reduced while chaperone capacity rises. If stress remains unresolved, prolonged UPR signaling can drive apoptosis.

CFTR preview: the common F508del protein has residual channel capability but misfolds enough to be retained/degraded by ER quality control, so trafficking failure becomes the disease mechanism.

Slides 28-29, 32

Coat proteins make vesicle budding energetically favorable and sort cargo

LO 5

Coat	Classic route	High-yield cue
COPII	ER → Golgi	Forward / anterograde secretory traffic
COPI	Golgi → ER and intra-Golgi retrieval	Retrograde / recycling
Clathrin	Plasma membrane endocytosis; Golgi/endosomal routes	Adaptor-mediated cargo concentration

Clathrin mechanics: heavy- and light-chain units assemble into triskelions and lattices; adaptor proteins link cargo to coat. BAR-domain proteins help bend membranes, and **dynammin-GTP** mediates scission at coated pits.

Slides 30-31

Rab proteins specify the target; SNAREs drive membrane fusion

LO 3

vesicle Rab

->

target Rab effector

->

v-SNARE pairs with t-SNARE

->

SNARE zippering expels water

->

fusion

After fusion, **NSF** uses ATP to disassemble the tightly wound SNARE complex so components can be recycled.

Docking vs fusion: Rab/Rab-effector provides identity and docking specificity; complementary SNAREs provide the mechanical fusion step.

Slides 32-34

Golgi sorting sends lysosomal enzymes by the mannose-6-phosphate pathway

LO 5

lysosomal hydrolase signal patch

->

GlcNAc phosphotransferase

->

M6P tag

->

M6P receptor in trans-Golgi

->

late endosome/lysosome

Disease	Defect	Consequence
Hurler disease	Specific lysosomal enzyme deficiency	GAG degradation impaired
I-cell disease	GlcNAc phosphotransferase deficiency	M6P tagging broadly fails; many lysosomal hydrolases are mis-sorted

Retrieval is essential: traffic is not one-way. ER residents that escape can be retrieved from Golgi, and receptors such as the M6P receptor must recycle.

Slides 35-39

Constitutive secretion is default; regulated secretion waits for a stimulus

LO 4-5

Pathway	Behavior	Example
Constitutive	Continuous/default delivery to membrane or extracellular space	General membrane renewal and ongoing secretion
Regulated	Cargo stored in secretory vesicles until a signal arrives	Neurotransmitter release; mast-cell granule release

Cystic fibrosis trafficking logic

CFTR is an ATP-gated chloride/bicarbonate channel that must traverse ER and Golgi to reach the apical membrane. **F508del** disrupts folding; ER quality control removes much of the mutant protein before it reaches the surface. Correctors improve folding/trafficking, while potentiators improve channel function.

Slides 40-45

Endocytosis brings cargo inward and uses pH to unload receptors

LO 5

As cargo moves plasma membrane → early endosome → late endosome → lysosome, pH falls. Many receptors bind ligand at neutral pH and release it after acidification; many receptors then recycle to the cell surface.

LDL

LDL receptor recruits adaptor + clathrin. Dynamin pinches off the vesicle. LDL is released in endosome; receptor recycles; cholesterol reaches lysosome. Defects can cause familial hypercholesterolemia.

EGF

EGF receptor is intentionally degraded after internalization, helping terminate a potent growth signal.

Pathogens/toxins

Viruses and toxins can exploit receptor-mediated endocytosis. The deck uses SARS-CoV-2 and botulinum toxin as trafficking examples.

Receptor fate is cargo-specific: LDL receptor is recycled; EGFR is often sent for degradation.

Slides 46-47

Phagocytosis and autophagy route large cargo to lysosomal degradation

LO 5

Phagocytosis

Specialized immune cells such as neutrophils and macrophages ingest large particles/pathogens, often using antibody or complement opsonization as recognition signals.

Autophagy

Cellular components are enclosed and ultimately delivered to lysosomes. It rises with stress/nutrient limitation but also contributes to normal basal housekeeping.

Exam traps

Protect route, coat, and disease relationships

All LOs

Trap 1: COPII = ER to Golgi; COPI = Golgi back to ER.

Trap 2: Rab specifies docking; SNARE pairing drives fusion.

Trap 3: Dynamin is a GTPase used in scission of clathrin-coated pits.

Trap 4: I-cell disease is a sorting/tagging defect, not simultaneous mutation of every hydrolase gene.

Trap 5: F508del CFTR is primarily a folding/trafficking problem before it is a surface-channel problem.

30-SECOND FINISH

Signal sequence = zip code. SRP docks translating ribosome at ER. Misfolded ER proteins go to ubiquitin/proteasome; overload triggers UPR. COPII out, COPI back, clathrin inward. Rab docks, SNARE fuses. M6P sends hydrolases to lysosome. Constitutive = default; regulated = stimulus.